

ComicVerseAI

AI-Powered Automated Comic Story & Visual Content Generation Platform

Project Name: ComicVerseAI
Team Members: P V Poornima, R Sandhiya, S Prathistha, Kiruthika K

PHASE 1: BRAINSTORMING & IDEA PRIORITIZATION

Step 1: Brainstorm and Idea Listing

Each team member listed multiple ideas and suggestions without judgment to address challenges in automated story generation, visual panel creation, and seamless AI comic production.

S.No	Team Member	Idea / Suggestion	Category
1	P V Poornima	Automated script generator using Gemini Flash for structured comic panel descriptions.	Backend AI Core
2	R Sandhiya	Stable Diffusion integration for consistent multi-panel visual art generation.	Image Generation
3	S Prathistha	Interactive web UI with character prompt customization and layout previews.	Frontend & UX
4	K Kiruthika	Automated PDF export pipeline with speech bubble layouts and custom styling.	Export Engine

PHASE 1: IDEA PRIORITIZATION & PROBLEM FORMULATION

Step 2: Idea Prioritization Matrix

Evaluating candidate features and technical paths based on feasibility and strategic impact for the ComicVerseAI platform.

Final Idea	Feasibility	Importance	Priority	Selected
AI Script & Storyboard Generator (Gemini)	High	High	P1	Yes
Stable Diffusion Comic Panel Generator	High	High	P1	Yes
FastAPI Web Interface & Panel Previewer	High	High	P1	Yes
Automated High-Resolution PDF Exporter	Medium	High	P2	Yes
3D Dynamic Camera View Renderer	Low	Medium	P3	No

PHASE 2: CUSTOMER PROBLEM STATEMENT

Understanding user requirements to build an intuitive, end-to-end comic generation experience.

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	An independent story writer or content creator	Transform written story concepts into visual comic strips	Lacking professional illustration skills and manual design software expertise	Hiring artists is expensive and manual drawing takes significant time	Frustrated by inability to visualize and share creative stories effectively

PHASE 3: EMPATHY MAP

Analyzing target user behaviors, motivations, and expectations across four key perspectives to drive customer-centric feature design.

PERSONA: Independent Comic Creator / Digital Storyteller

SAYS

- "I have great story ideas, but I can't draw comic panels."
- "I need a fast tool to generate artwork that matches my script."
- "Hiring illustrators is too expensive for small indie stories."
- "I want to publish my comics online in standard PDF format."

THINKS

- "AI should be able to generate consistent visual scenes from my text."
- "Will the generated character stay"
- "How can I quickly iterate on dialogue and story flow?"

DOES

- Writes scripts and narrative prompts in text editors.
- Searches for affordable graphic generation tools and assets.
- Manually arranges storyboards or attempts basic visual edits.
- Shares early story drafts with online communities for feedback.

FEELS

- **Overwhelmed** by the complexity of traditional drawing tools.
- **Excited** by the possibilities of generative AI art.
- **Eager** to see story ideas converted into visuals instantly.
- **Determined** to produce high-quality indie digital comics.

PHASE 4: SYSTEM ARCHITECTURE & REQUIREMENTS

Core Technology Stack

Component	Technology Selected	Purpose & Description
Backend Framework	FastAPI (Python)	Asynchronous REST API routing, task management, and request processing.
Story Generation AI	Google Gemini Flash	Parsing prompt input into structured multi-panel script & scene prompts.
Image Generation Engine	Stable Diffusion API	Transforming scene text descriptions into comic panel artwork.
Frontend Interface	HTML5, CSS3, JavaScript	Interactive web interface for user prompts, panel views, and edits.
Export Engine	WeasyPrint / ReportLab	Compiling image panels and script text into formatted PDF books.

Directory Architecture Plan

- `app/`: Core application backend logic (`main.py`, `routes.py`, `gemini_flash.py`).
- `templates/`: Web UI layout components (`index.html`).
- `static/`: Generated panel output (`panels/`), exports (`exports/`), and styles.
- `requirements.txt`: Python dependencies for local and cloud execution.

PHASE 5: DEVELOPMENT & TESTING PLAN

Validation & Quality Assurance Strategy

Test Area	Test Objective	Expected Outcome
Script Parsing	Validate Gemini response parsing under varied prompt length	Clean JSON output containing panel dialogues and visual prompts
Image Generation	Verify Stable Diffusion image output speed and resolution	512x512 / 1024x1024 crisp visual panel generation
Pipeline Integration	End-to-end user flow test from prompt to panel render	Seamless execution within local Uvicorn environment
PDF Compilation	Export test with multi-page comic layouts	Cleanly formatted PDF ready for print and sharing

Project Summary & Delivery

ComicVerseAI streamlines digital comic creation by pairing generative text models with diffusion-based rendering. The complete repository source code, environment templates, and documentation are hosted on GitHub for version control and community deployment.